

Scalar QED: $A_3(\varphi\varphi^*) = t \rightarrow C^{\pm 2}$

$$A_2(1^{h_1} 2^{h_2} 3^{h_3}) = C \langle 12 \rangle^{h_3 - h_1 - h_2} \langle 13 \rangle^{h_2 - h_1 - h_3} \langle 23 \rangle^{h_1 - h_2 - h_3}$$

$$A_3(\varphi\varphi^*\gamma^+) = C \frac{\langle 12 \rangle}{\langle 13 \rangle \langle 23 \rangle} \sim (\text{mass})^{-1}$$

comes from $ieA^\mu(\partial_\mu\varphi^*\varphi - \varphi^*\partial_\mu\varphi)$ term

$$A_3(\varphi\varphi^*\gamma^+) = C \frac{[13][23]}{[12]} \sim (\text{mass})^2$$

$$A_3(\varphi\varphi^*\gamma^-) = C \frac{\langle 13 \rangle \langle 23 \rangle}{\langle 12 \rangle}$$

$$\Rightarrow A_3 = i(p_1 - p_2)^\mu \epsilon_\mu^{(-)}(p_3)$$

$$= -i(p_1 - p_2)^\mu \langle 3 | \gamma^\mu | q \rangle$$

$$= -\frac{i}{\sqrt{2}[3q]} (\langle 3 | 1 | q \rangle - \langle 3 | 2 | q \rangle)$$

$$= \frac{-i}{\sqrt{2}[3q]} (\langle 31 \rangle [q] - \langle 32 \rangle [2q])$$

Set $q = 1$:

$$= \frac{-i}{\sqrt{2}[31]} \langle 32 \rangle [21]$$

$$(1) \rightarrow [11] + [2] [212] + [3] [311] = 0$$

$$\langle 12 \rangle [12] = -\langle 13 \rangle [13]$$

$$[21] = -\frac{\langle 13 \rangle}{\langle 12 \rangle}$$

$$[31] = \frac{\langle 12 \rangle}{\langle 13 \rangle}$$

$$\rightarrow = \frac{i}{\sqrt{2}} \frac{\langle 32 \rangle \langle 13 \rangle}{\langle 12 \rangle} \quad \checkmark$$

$$A_3(\varphi\varphi^*\gamma^-) = ie \frac{\langle 23 \rangle \langle 31 \rangle}{\langle 12 \rangle}$$

* Use a $[u, 13]$ shift on $A_4(\varphi\varphi^*\gamma^+\gamma^-)$

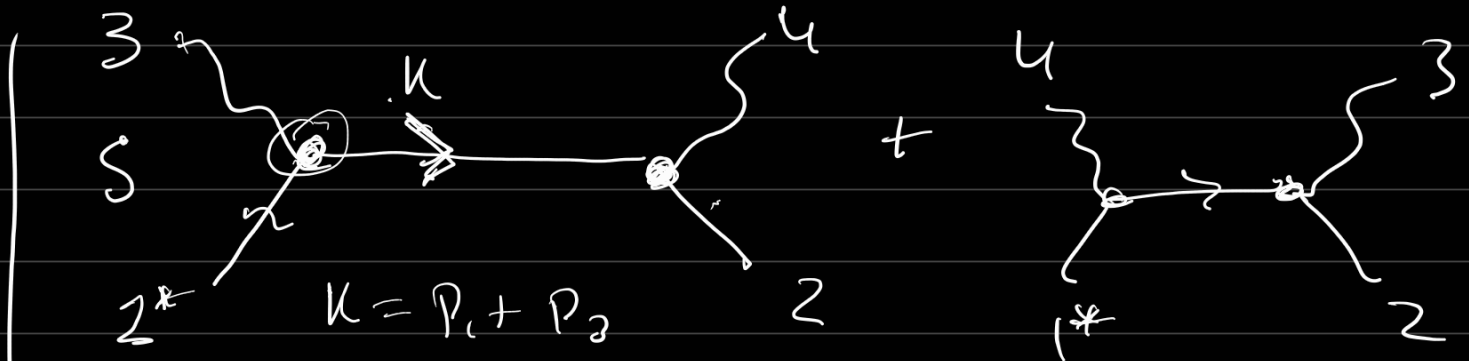
Will have tree-level terms from $e^2 A^\mu A_\mu \varphi^* \varphi$

i.e. m_γ^2



* 4-vertex: $V(p_1, p_2, p_3, p_4) = -ie^2 \eta_{\mu\nu} \epsilon_3^{(+)\mu} \epsilon_4^{(-)\nu}$

$$\langle \phi_1 \phi_2^* A_3^\mu A_4^\nu \rangle = -ie^2 \int d^4\omega d^4z (A_3^\mu \partial_\mu \phi^* \phi)^2 \phi_1 \phi_2^* A_3^\mu A_4^\nu$$



$$ie \epsilon_3^{(+)\mu}(p_3) \left(\frac{(p_1)^\mu}{(p_1 + p_3)^2} \right) ie \epsilon_4^{(-)\nu}(p_4) (p_2)^\nu$$

$$\langle A_2^\sigma \partial_\sigma \phi_2^* \phi_2 \phi_1 \phi_2^* A_3^\mu A_4^\nu A_1^\lambda \partial_\lambda \phi A \rangle$$

* Only V contributes:

$$A_4 = -ie^2 \left(\frac{\langle q_3 | \gamma^\mu | 3 \rangle}{\sqrt{2} \langle q_3 3 \rangle} \right) \left(\frac{\langle 4 | \gamma_\mu | q_4 \rangle}{\sqrt{2} [q_4]} \right)$$

$$= -ie^2 \frac{\langle q_3 4 \rangle [3 q_4]}{\langle q_3 3 \rangle [q_4 4]} \quad \begin{array}{l} q_3 = 1 \\ q_4 = 2 \end{array}$$

$$= 2ie^2 \frac{\langle 1 4 \rangle [3 2]}{\langle 1 3 \rangle [3 4]} *$$

$$* S: -ie^2 \left(\frac{(-1) \langle 3 1 \rangle [1 q_3]}{\sqrt{2} [q_3 3]} \right) \left(\frac{(-1) \langle q_4 2 \rangle [2 4]}{\sqrt{2} \langle q_4 4 \rangle} \right)$$

$$= e^2 \frac{[1 q_3] \langle q_4 2 \rangle [2 4]}{[q_3 3] [1 3] \langle q_4 4 \rangle}$$

$$q_3 = 1, q_4 = 2$$

$$= \frac{[1 2] \langle 3 2 \rangle [2 4]}{[2 3] [1 3] \langle 3 4 \rangle} \rightarrow 0$$

$$\langle 1 3 \rangle [3 1 2] + \langle 1 4 \rangle [4 1 2] = 0$$

$$[2 4] = -\langle 1 3 \rangle$$

$$\begin{aligned} & [23] \langle 21 \rangle \\ & \rightarrow + \frac{[12] \langle 23 \rangle \langle 13 \rangle}{\langle 14 \rangle [13] \langle 34 \rangle} \quad * \end{aligned} \quad \left| \begin{array}{l} ie(p_{out} - p_{in}) \end{array} \right.$$

* Second graph: $e^2 \frac{[2q_3] \langle q_4 1 \rangle [14]}{[q_3 3] [23] \langle q_4 4 \rangle}$

$$q_3 = 1, q_4 = 2$$

$$= e^2 \frac{[21] \langle 21 \rangle [14]}{[13] [23] \langle 24 \rangle}$$

$$\therefore A_u = e^2 [23] \langle 34 \rangle \left(\frac{[12] \langle 32 \rangle [24]}{[13]} + \langle 31 \rangle \right)$$

$$[12]$$

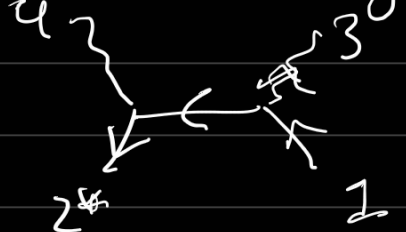
* $\frac{[12] \langle 32 \rangle [24]}{[23] [13] \langle 34 \rangle}$



* $ie(p_1 + p_3 - (-p_1))^\mu \epsilon_\mu^{(+)}(p_3)$

$$\times \frac{1}{(p_1 + p_3)^2} ie(p_2 + p_4 - (-p_2))^\nu \epsilon_\nu^{(-)}(p_4)$$

= 0 via gauge choice



$$\frac{1}{ie(p_2 + p_4 - (-p_2))^\nu \epsilon_\nu^{(-)}(p_4) \langle 24 \rangle [24]} \frac{1}{ie(p_1 + p_3 - (-p_1))^\mu \epsilon_\mu^{(+)}(p_3)}$$

$$ie^2 (2p_2 + p_4)^\mu \frac{\langle 4 | \gamma_\mu | 1 \rangle}{\sqrt{2} [14] \langle 24 \rangle [24]} \frac{1}{(2p_2 + p_4)^\nu \langle 3 | \gamma_\nu | 2 \rangle \sqrt{2} [23]}$$

$$= ie^2 \frac{\langle 42 \rangle [21] \langle 34 \rangle [42]}{[14] \langle 24 \rangle [24] [23]}$$

$$= ie^2 \frac{[21] \langle 34 \rangle}{[14] [23]}$$

$$\langle 43 \rangle [32] + \langle 41 \rangle [12] = 0$$

$$= - \frac{[21] \langle 34 \rangle}{[14] [23]}$$

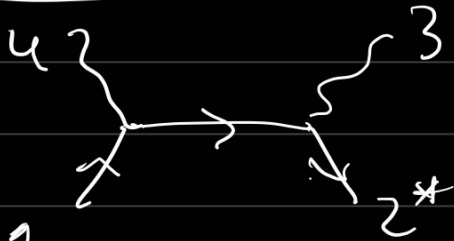
$$\frac{[23]}{[22]} = -\frac{[43]}{[41]}$$

$$\Rightarrow ie^2 \frac{[43][34]}{[41][14]}$$

$$\begin{aligned} & \frac{[34]^2 ie^2}{[41][14]} + 2ie^2 \frac{[14][32]}{[13][34]} \\ &= \frac{[13][34][34]^2 - 2[14]^2[14][32]}{[14][14][13][34]} \end{aligned}$$

2-vertex:

$$\begin{aligned} q_3 &= 1 \\ q_u &= 2 \end{aligned}$$



$$\begin{aligned} & ie (\epsilon^\mu(p_u))_\mu (p_u + 2p_1)^\mu \left(\frac{1}{[14][14]} \right) \epsilon^\nu(p_3)^\mu (p_3 + 2p_2)_\mu \\ &= e^2 \left(\frac{[41][12]}{\sqrt{2}[24][14]} \frac{1}{[11][13]} \right) (p_u + 2p_1)_\mu (p_3 + 2p_2)_\nu \\ &= 2e^2 \left(\frac{[41][12][12][23]}{[24][14][13]} \right) \end{aligned}$$

$$= 2e^2 \left(\frac{[12][12][23]}{[24][14][13]} \right)$$

$$\begin{aligned} & \left[\begin{aligned} [12][21] + [13][31] + [14][41] &= 0 \\ [12][23] + [14][43] &= 0 \\ \frac{[12]}{[14]} &= -\frac{[34]}{[23]} \end{aligned} \right] \\ & \Rightarrow 2e^2 \left(\frac{[12][34][14]}{[23][23][13]} \right) \end{aligned}$$

$$\begin{aligned} & -[34][41] = [31] - (1+2+3)[1] \\ & = [32][21] \end{aligned}$$

$$\Rightarrow 2e^2 \left(\frac{[12][32][21]}{[23][23][13]} \right) = \frac{2e^2 [12][21]}{[23][13]}$$

$$[21][11] + [23][31] + [24][41] = 0$$

$$[21] \langle 14 \rangle + [23] \langle 34 \rangle = 0$$

$$\frac{[21]}{[23]} = - \frac{\langle 34 \rangle}{\langle 14 \rangle}$$

$$= 2e^2 \frac{\langle 12 \rangle \langle 34 \rangle}{\langle 14 \rangle \langle 13 \rangle}$$



$$2e^2 \left(\frac{\langle 12 \rangle \langle 34 \rangle}{\langle 14 \rangle \langle 13 \rangle} \right) + \frac{\langle 14 \rangle [32]}{\langle 13 \rangle [34]} = A_4(\varphi \varphi^* \gamma^- \gamma^+)$$

* Proof using BCFW: A.

$$A_n = \sum_{\text{diagrams } I} \tilde{A}_L(z_I) \frac{1}{p_I^2} \hat{A}_R(z_I)$$

$$= \sum_{\substack{I=14 \\ h_I = +}} \hat{A}_3[\varphi, \hat{p}_I^{h_I} \gamma^-] \frac{1}{p_I^2} A_3[-\hat{p}_I^{h_I}, \varphi^* \gamma^+]$$

$$= \hat{A}_3[\varphi, \hat{p}_{14}^+ \gamma^-] \frac{1}{p_{14}^2} \hat{A}_3[-\hat{p}_{14}^-, \varphi^*, \gamma^+]$$

$$* A_3[\varphi \varphi^* \gamma^+] = \frac{\langle 23 \rangle \langle 31 \rangle}{\langle 12 \rangle}$$

little group scaling to find $A_3[\varphi \varphi^* \gamma^+]$

$$A_3[\varphi \varphi^* \gamma^+] = [12] [13]^{-2} [23]^{-1}$$

under $U(1) \cong SO(2)$:

$$A_3 \rightarrow t^{-2} A_3$$

A_3 here has mass dimension $(\text{mass})^{-2}$

$\Rightarrow$ should be:

$$ie \epsilon_{\mu\nu\rho} (p_2 - p_1)^\mu \sim ie \frac{\langle 31 \rangle (p_2 - p_1)_\rho}{\sqrt{2} [43]}$$

$$= -ie \frac{\langle 32 \rangle [24] - \langle 31 \rangle [14]}{\sqrt{2} [43]} \quad q = 1$$



$$= i e \frac{\langle 32 \rangle [12]}{\sqrt{2} [13]} \left| \begin{array}{l} [12] \langle 21 \rangle + [13] \langle 31 \rangle = 0 \\ [12] = \frac{\langle 31 \rangle}{\langle 21 \rangle} \end{array} \right.$$

$$= i e \frac{\langle 23 \rangle \langle 13 \rangle}{\langle 21 \rangle} = A_3 [\varphi \varphi^* \gamma^*]$$

$$\therefore \hat{A}_\mu [\varphi \varphi^* \gamma^* \gamma^-] = e^2 \frac{\langle 23 \rangle \langle 13 \rangle}{\langle 21 \rangle} \frac{1}{\langle 14 \rangle [14]} \frac{[23] [13]}{[21]}$$

$$= \frac{\langle 23 \rangle \langle 13 \rangle [23] [13]}{\langle 14 \rangle [14] [21] \langle 21 \rangle}$$

$$\left\{ \begin{array}{l} \frac{[13]}{[14]} = - \frac{\langle 42 \rangle}{\langle 32 \rangle} \\ \Rightarrow \frac{\langle 23 \rangle \langle 13 \rangle \langle 24 \rangle [23]}{\langle 32 \rangle \langle 14 \rangle [21] \langle 21 \rangle} = \frac{\langle 13 \rangle \langle 24 \rangle [23]}{\langle 14 \rangle [12] \langle 21 \rangle} \end{array} \right.$$

$$\left\{ \begin{array}{l} \frac{[23]}{[12]} = - \frac{[23]}{[21]} = \frac{\langle 14 \rangle}{\langle 34 \rangle} \\ \Rightarrow \frac{\langle 13 \rangle \langle 24 \rangle \langle 14 \rangle}{\langle 14 \rangle \langle 21 \rangle \langle 34 \rangle} = \frac{\langle 24 \rangle \langle 13 \rangle}{\langle 21 \rangle \langle 34 \rangle} \end{array} \right.$$

* BCFW for Gravitational Scattering:

3-graviton scattering: $g_{\mu\nu\rho} = \eta_{\mu\nu} + k_\mu k_\nu$

$\rightarrow$ work with linearized gravity

$$A_3(h_1^{\mu}, h_2^{\mu}, h_3^{\mu})$$

$$A_3 = c \langle 12 \rangle^{h_3 - h_1 - h_2} \langle 13 \rangle^{h_2 - h_1 - h_3} \langle 23 \rangle^{h_1 - h_2 - h_3}$$

For $(---)$:

$$A_3 = \langle 12 \rangle^2 \langle 13 \rangle^2 \langle 23 \rangle^2$$

Under $U(1)$: (on 1)

$$A_3 \rightarrow t^{-4} A_3$$

so A_3 is allowed by the little group

For $(++-)$:

$$A_3 \sim \langle 12 \rangle^2 \langle 13 \rangle^2 \langle 23 \rangle^2$$

under $U(1)$:

$$A_3 \rightarrow t^{-4} A_3 \text{ (should be } t^{+4})$$

so

$$\rightarrow A_3[+++] = [12]^2 [13]^2 [23]^2$$

* Mixed:

$$A_3[-++]\overset{-6}{=} \langle 12 \rangle^2 \langle 13 \rangle^2 \langle 23 \rangle^2 = \frac{\langle 12 \rangle^2 \langle 13 \rangle^2}{\langle 23 \rangle^6}$$

Under $U(1)$ (for 3):

$$A_3 \rightarrow t^{-4} A_3$$

(for 1)

$$A_3 \rightarrow t^{+4} A_3 \quad \text{or}$$

* For $A_3[- - +]$:

$$A_3[- - +] = \langle 12 \rangle^6 \langle 13 \rangle^{-2} \langle 23 \rangle^{-2}$$

Under $U(1)$ (for 3):

$$A_3 \rightarrow t^{-4} A_3$$

$$\text{(for 2)} \quad A_3 \rightarrow t^{+4} A_3:$$

Interesting note:

$$A_3[h_1^- h_2^- h_3^-] = (\langle 12 \rangle \langle 13 \rangle \langle 23 \rangle)^2$$

$$\text{and } A[h_1^- h_2^- h_3^+] = \left(\frac{\langle 12 \rangle^3}{\langle 13 \rangle \langle 23 \rangle} \right)^2$$

(gravity)

→ For Yang-Mills: Gluon scattering (massless)

$$A_3^{\text{YM}}[g_1^- g_2^- g_3^+] = \frac{\langle 12 \rangle^3}{\langle 13 \rangle \langle 23 \rangle} \quad \left[A_n^{\text{gr}} \equiv M_n \right]$$

⇒ This looks like:

$$\text{* } \boxed{\text{(gravity)}^{\text{(linear)}} = (\text{Yang Mills})^2} \text{ *}$$

* That's cool.

$$A_3^{grav} [h_1^-, h_2^-, h_3^+] = \left(A_3^{YM} [g^-, g^-, g^+] \right)^2$$

* Also true for

$$A_3^{YM} [g^-, g^-, g^-] = \langle 12 \rangle \langle 13 \rangle \langle 23 \rangle = \sqrt{A_3^{grav} [h^-, h^-, h^-]}$$

→ Try with 4 pt: $M_4 [1^- 2^- 3^+ 4^+]$

[1, 2] - BCFW shift Bern

$$M_4 [1^- 2^- 3^+ 4^+] = \sum_{h_I = \pm} \hat{A}_3 [1^- \hat{p}_I^+ 4^+] \frac{1}{\hat{p}_I^2} \hat{A}_3 [-\hat{p}_I^-, 2^- 3^+]$$

$$= \hat{A}_3 [1^- \hat{p}_I^+ 4^+] \frac{1}{\hat{p}_I^2} \hat{A}_3 [-\hat{p}_I^-, 2^- 3^+] \quad \textcircled{a}$$

$$+ \hat{A}_3 [1^- \hat{p}_I^- 4^+] \frac{1}{\hat{p}_I^2} \hat{A}_3 [-\hat{p}_I^+, 2^- 3^+] \quad \textcircled{b}$$

$$\textcircled{a} \hat{A} [-++]\frac{1}{\hat{p}_I^2} \hat{A}_3 [+ - +],$$

→ Use above "duality"

$$M_3 [-++]=\frac{[12]^6}{[23]^2[31]^2}$$

$$M_3 [+ - +]=\frac{[12]^+ [13]^{-3} [23]}{\left(\frac{[12][23]}{[13]^3}\right)^2}$$

does this scale properly?

$$M_3 \rightarrow t^{-4} M_3$$

$$\Rightarrow \frac{[\hat{1}\hat{p}]^6}{[\hat{p}\hat{3}]^2[\hat{3}\hat{1}]^2} \frac{1}{\langle \hat{1}\hat{4} \rangle [\hat{4}\hat{1}]} \left(\frac{[\hat{p}\hat{2}]^2 [\hat{2}\hat{3}]^2}{[\hat{p}\hat{3}]^6} \right)$$

$$= \left(\frac{[\hat{1}\hat{p}]^6 [\hat{p}\hat{2}]^2 [\hat{2}\hat{3}]}{[\hat{p}\hat{3}]^8 [\hat{3}\hat{1}]^2 \langle \hat{1}\hat{4} \rangle [\hat{4}\hat{1}]} \right) \frac{\langle \hat{p}\hat{2} \rangle^8}{\langle \hat{p}\hat{3} \rangle^8}$$

$$[\hat{3}\hat{p}]\langle \hat{p}\hat{2} \rangle = -[\hat{3}|\hat{1}+\hat{4}|\hat{2}\rangle$$

$$= [\hat{3}\hat{1}]\langle \hat{1}\hat{2} \rangle + [\hat{3}\hat{4}]\langle \hat{4}\hat{2} \rangle$$

⇒ This diagram is zero.

$\textcircled{a} = 0$

$$\textcircled{b} A_3 [\hat{p}_1^+ \hat{p}_1^- 4^+] \frac{1}{p_1^2} A_3 [-\hat{p}_1^+ \hat{z}^- 3^+]$$

$$\frac{\langle \hat{1} \hat{p} \rangle^6}{\langle \hat{1} 4 \rangle^2 \langle \hat{p} 4 \rangle^2} \left(\frac{1}{\langle 14 \rangle [14]} \right) \frac{[\hat{p} \hat{z}]^2 [\hat{z} 3]^2}{[\hat{p} 3]^6} \quad \text{BRUH}$$

$$(i) \langle \hat{1} \hat{p} \rangle [\hat{p} \hat{z}] = -\langle \hat{1} | (\hat{1} + 4 | \hat{z} \rangle$$

$$= \langle 14 \rangle [4 \hat{z}]$$

$$(ii) [\hat{z} \hat{p}] \langle \hat{p} 4 \rangle = [\hat{z} | \hat{1} \rangle \langle \hat{1} 4 \rangle$$

$$= [\hat{z} \hat{1}] \langle 14 \rangle$$

$$\Rightarrow \frac{\langle \hat{1} \hat{p} \rangle^4 \langle 14 \rangle^2 [\hat{z} 4]^2 [\hat{z} 3]^2}{\langle 14 \rangle^2 \langle 14 \rangle^2 [\hat{z} 1]^2 \langle 14 \rangle [14] [\hat{p} 3]^4}$$

$$= \frac{[\hat{z} 4]^2 [\hat{z} 3]^2}{[\hat{z} 1]^2 \langle 14 \rangle [14]} \left(\frac{\langle \hat{1} \hat{p} \rangle^4}{[\hat{p} 3]^4} \right)$$

$$[\hat{z} 1] \langle \hat{1} 4 \rangle + [\hat{z} \hat{z}] \langle \hat{z} 4 \rangle$$

$$\therefore \left(\frac{[\hat{z} 2]}{[\hat{z} 1]} \right)^2 = \left(\frac{\langle \hat{z} 4 \rangle}{\langle 14 \rangle} \right)^2$$

$$= \frac{[\hat{z} 4]^2 \langle \hat{z} 4 \rangle^2}{\langle 14 \rangle^2 \langle 14 \rangle [\hat{1} 4]} \left(\frac{\langle \hat{1} \hat{p} \rangle^4}{[\hat{p} 3]^4} \right)$$

$$-\langle \hat{1} 4 \rangle [4 \hat{z}] = \langle \hat{1} | [3 + \hat{z} | \hat{1}]$$

$$= -\langle \hat{1} 3 \rangle [\hat{z} \hat{1}] - \langle 12 \rangle [\hat{z} 1]$$

$$\Rightarrow \langle \hat{p} | \hat{1} \rangle [\hat{1} | + \langle \hat{p} | \hat{z} \rangle [\hat{z} | + \langle \hat{p} | \dots$$

$$0 = \langle \hat{p} 1 \rangle [\hat{1} \hat{p}] + \langle \hat{p} 2 \rangle [\hat{z} \hat{p}] + \langle \hat{p} 3 \rangle [\hat{p} \hat{z}] + \dots$$

$$\textcircled{a} \langle \hat{1} \hat{p} \rangle [\hat{p} \hat{z}] = \langle 14 \rangle [4 \hat{z}]$$

$$\textcircled{b} [\hat{z} \hat{p}] \langle \hat{p} \hat{z} \rangle = -[\hat{z} 3] \langle \hat{z} 3 \rangle$$

$$\textcircled{c} [\hat{z} \hat{p}] \langle \hat{p} 3 \rangle = -[\hat{z} \hat{z}] \langle \hat{z} 3 \rangle$$

$$\textcircled{d} [4 \hat{p}] \langle \hat{p} 4 \rangle = [4 \hat{1}] \langle \hat{z} 4 \rangle$$

$$\ast \frac{[\hat{z} 4]^2 \langle \hat{z} 4 \rangle^2 \langle \hat{1} \hat{p} \rangle^4}{\langle 14 \rangle^2 \langle 14 \rangle [\hat{1} 4] [\hat{p} 3]^4} \frac{[\hat{p} 2]^4}{[\hat{p} 2]}$$

$$\frac{\langle \hat{z} \hat{p} \rangle^6}{\langle \hat{z} 4 \rangle^2 \langle \hat{p} 4 \rangle^2} \left(\frac{1}{\langle 14 \rangle [14]} \right) \left(\frac{[\hat{p} 3]^6}{[\hat{p} 2]^2 [\hat{z} 3]^2} \right)$$

$$(i) \langle \hat{1} \hat{p} \rangle [\hat{p} 3] = \langle \hat{z} 4 \rangle [4 3] = \langle 14 \rangle [4 3]$$

$$(b) \langle 4\hat{\phi} \rangle [\hat{\phi}^2] = \langle 4\hat{\pi} \rangle [\hat{\pi}^2] = \langle 14 \rangle [12]$$

$$\rightarrow \frac{\langle 14 \rangle^6 [43]^6}{\langle 14 \rangle \langle 14 \rangle^2 [12]^2 \langle 14 \rangle [14] [23]^2} \\ = \frac{\langle 14 \rangle^2 [34]^6}{[12]^2 [14] [23]}$$

$$\left(\begin{aligned} \langle 14 \rangle [43] &= -\langle 12 \rangle [23] \\ \rightarrow \frac{\langle 12 \rangle^2 [23]^2 [34]^2}{[12]^2 [14] [23]} \end{aligned} \right)$$

$$\left(\begin{aligned} \left(\frac{[23]}{[21]} \right)^2 &= \left(\frac{\langle 14 \rangle}{\langle 34 \rangle} \right)^2 \\ \rightarrow \frac{\langle 12 \rangle^2 \langle 14 \rangle^2 [34]^2}{[21] \langle 34 \rangle [14] [23]} \end{aligned} \right)$$

$$\left(\begin{aligned} [23] \langle 34 \rangle &= -[21] \langle 14 \rangle \\ \rightarrow \frac{\langle 12 \rangle^2 \langle 14 \rangle [34]^2}{[21] \langle 34 \rangle [14] [23]} \end{aligned} \right)$$

$$\left(\begin{aligned} [34] \langle 41 \rangle + [32] \langle 21 \rangle + [31] \langle 11 \rangle &= 0 \\ [34] \langle 42 \rangle &= +[13] \langle 12 \rangle \\ \rightarrow \frac{\langle 12 \rangle^2 \langle 14 \rangle [13]^2 \langle 12 \rangle^2}{\langle 34 \rangle^2 [14] \langle 24 \rangle^2 [23]} \end{aligned} \right)$$

$$\mu_u(1^- 2^- 3^+ 4^+) = \frac{\langle 12 \rangle^8 [12]}{\langle 12 \rangle \langle 13 \rangle \langle 14 \rangle \langle 23 \rangle \langle 24 \rangle \langle 34 \rangle^2}$$

$$\left(\begin{aligned} [12] \langle 21 \rangle + [13] \langle 31 \rangle + [14] \langle 41 \rangle &= 0 \\ [12] \langle 23 \rangle + [14] \langle 43 \rangle \\ \rightarrow [12] &= -\frac{[14] \langle 43 \rangle}{\langle 23 \rangle} \end{aligned} \right)$$

also equal to

$$\mu_u = \frac{\langle 12 \rangle^6 [34]^2}{\langle 12 \rangle [12] \langle 23 \rangle [23] \langle 13 \rangle [12]}$$

$$\langle 12 \rangle \langle 13 \rangle \langle 23 \rangle \langle 14 \rangle \langle 24 \rangle \langle 34 \rangle$$

$$\text{u-channel: } A_3[\hat{1}^- \hat{p}_u^+ 4^+] \left(\frac{1}{s_u} \right) A[-\hat{p}^+ \hat{2}^- 3^+]$$

$$\mu_3[-+-] = \frac{[23]^6}{[12]^2 [13]^2}$$

$$\mu_3[--+] = \frac{\langle 12 \rangle^6}{\langle 13 \rangle \langle 23 \rangle}$$

$$\mu_3[+-+] = \frac{[13]^6}{[21]^2 [23]^2}$$

$$(-1) \left(\frac{[14]^6}{[\hat{p}1]^2 [\hat{p}4]^2} \right) \left(\frac{1}{\langle 14 \rangle [14]} \right) \left(\frac{\langle \hat{p}2 \rangle^6}{\langle \hat{p}3 \rangle^2 \langle \hat{2}3 \rangle^2} \right)$$

$$= (-1) \frac{[14]^6}{[\hat{p}1]^2 [\hat{p}4]^2} \frac{1}{\langle 14 \rangle [14]} \left(\frac{\langle \hat{p}2 \rangle^6}{\langle \hat{p}3 \rangle^2 \langle \hat{2}3 \rangle^2} \right)$$

$$\langle 3 \hat{p} \rangle [\hat{p}1] = \langle 34 \rangle [4 \hat{1}]$$

$$\frac{[14]^4 \langle \hat{p}2 \rangle^6}{[\hat{p}4]^2 \langle 34 \rangle^2 \langle 14 \rangle [14] \langle \hat{2}3 \rangle^2}$$

$$* \hat{p}^2 = \langle 14 \rangle [14] = 0 \quad [14] + 2[24] = 0$$

$$p \cdot [\hat{p}14] = 0$$

$$q \cdot \langle \hat{p}2 \rangle = 0$$

$$2z = \frac{[14]}{[24]}$$

$$\text{t-channel: } \hat{p}_t = \sum_i \hat{p}_i$$

$$|1\rangle [1] = |2\rangle ([1] + z[2])$$

$$|-\hat{p}\rangle [\hat{p}] = -|2\rangle [1] + |1\rangle [2] (z + 14)[4]$$

$$[\hat{p}] [\hat{p}4] = -[1][14] - [2][24] z$$

$$= -[1][14] + [2][14] z$$

$$= [14] (|2\rangle - |1\rangle)$$

$$* |\hat{p}\rangle = \frac{[14]}{[\hat{p}4]} (|2\rangle - |1\rangle)$$

$$1 \quad 1 \quad 1 \quad \frac{1}{s} \quad 11 \quad [\hat{p} - \hat{2} = 3^+] \quad \text{green circle}$$

$$A_3[\hat{I}^+ \hat{P}_I^- 4^+] \left(\frac{1}{P_{14}^2} \right) A_3[\hat{I}^- \hat{P}_I^+ 2^+] \quad (a)$$

$$+ A_3[\hat{I}^- \hat{P}_I^- 4^+] \left(\frac{1}{P_{14}^2} \right) A_3[\hat{I}^+ \hat{P}_I^+ 2^+] \quad (b)$$

$$= \frac{[\hat{P}_I 4]^6}{[\hat{I} 1]^2 [\hat{I} 3]^2} \left(\frac{1}{P_{14}^2} \right) \frac{(\hat{P}_I 2)^6}{[\hat{I} 3]^2 [\hat{I} 1]^2}$$

$$\therefore \langle \hat{I} 2 \rangle [\hat{P}_I 4] = \langle \hat{I} 1 \rangle [\hat{I} 4]$$

$$\langle \hat{I} 3 \rangle [\hat{P}_I 2] = \langle \hat{I} 4 \rangle [\hat{I} 1]$$

$$\frac{\langle \hat{I} 1 \rangle^6 [\hat{I} 4]^6}{[\hat{I} 4]^2 [\hat{I} 3]^2 [\hat{I} 2]^2 [\hat{I} 1]^2}$$

$$= \frac{[\hat{I} 4]^2 \langle \hat{I} 3 \rangle^2 [\hat{I} 2] \langle \hat{I} 3 \rangle^2}{[\hat{I} 4]^2 \langle \hat{I} 3 \rangle^2 [\hat{I} 2] \langle \hat{I} 3 \rangle^2}$$

$$(a) = \frac{\langle \hat{I} 2 \rangle^6 [\hat{I} 4]^4}{\langle \hat{I} 3 \rangle^2 [\hat{I} 2] \langle \hat{I} 3 \rangle^2} = 0$$

$$(b) = \frac{(\langle \hat{I} 1 \rangle)^6}{(\langle \hat{I} 4 \rangle^2 \langle \hat{I} 3 \rangle^2) \left(\frac{1}{\langle \hat{I} 4 \rangle [\hat{I} 4]} \right) \left(\frac{[\hat{P}_I 3]^6}{[\hat{I} 3]^2 [\hat{I} 2]^2} \right)}$$

$$\langle \hat{I} 1 \rangle [\hat{P}_I 3] = \langle \hat{I} 4 \rangle [\hat{I} 3]$$

$$\langle \hat{I} 4 \rangle [\hat{P}_I 2] = \langle \hat{I} 1 \rangle [\hat{I} 2]$$

$$= \frac{[\hat{I} 4]^6 [\hat{I} 3]^6}{[\hat{I} 4]^2 [\hat{I} 1]^2 [\hat{I} 2]^2 [\hat{I} 4] [\hat{I} 3]^2}$$

$$= \frac{[\hat{I} 4]^6 [\hat{I} 3]^6}{[\hat{I} 2]^2 [\hat{I} 4] [\hat{I} 3]^2}$$

$$\langle \hat{I} 2 \rangle [\hat{I} 3] + \langle \hat{I} 4 \rangle [\hat{I} 3] = 0$$

$$[\hat{I} 3] = - \langle \hat{I} 2 \rangle [\hat{I} 3]$$

$$\rightarrow \frac{\langle \hat{I} 4 \rangle \langle \hat{I} 2 \rangle^6 [\hat{I} 3]^6}{[\hat{I} 4]^6 [\hat{I} 2]^2 [\hat{I} 4] [\hat{I} 3]^2}$$

$$= \frac{\langle \hat{I} 2 \rangle^6 [\hat{I} 3]^4}{[\hat{I} 4]^5 [\hat{I} 2]^2 [\hat{I} 4]}$$

$$\left(\begin{array}{l} \bullet [\hat{I} 2] \langle \hat{I} 4 \rangle = - [\hat{I} 3] \langle \hat{I} 3 \rangle \\ \hookrightarrow \frac{\langle \hat{I} 2 \rangle^6 [\hat{I} 3]^2}{\langle \hat{I} 4 \rangle^3 \langle \hat{I} 3 \rangle^2 [\hat{I} 4]} \end{array} \right)$$

$$\frac{\langle \hat{I} 2 \rangle^6 [\hat{I} 3]^2}{\langle \hat{I} 4 \rangle^3 \langle \hat{I} 3 \rangle^2 [\hat{I} 4]}$$

$$\left(\begin{aligned} [21] \langle 21 \rangle + [23] \langle 31 \rangle + [24] \langle 41 \rangle &= 0 \\ [21] \langle 14 \rangle + [23] \langle 34 \rangle \\ [23] &= - \frac{[21] \langle 14 \rangle}{\langle 34 \rangle} \end{aligned} \right)$$

$$\rightarrow \frac{\langle 12 \rangle^6 [12]^2 \langle 14 \rangle^2}{\langle 14 \rangle^3 \langle 34 \rangle^4 [14]} = - \frac{\langle 12 \rangle^6 [12]^2}{\langle 14 \rangle \langle 34 \rangle^4 [14]}$$

$$\left(\begin{aligned} \langle 34 \rangle [41] &= \langle 32 \rangle [21] \\ \frac{\langle 12 \rangle^6 [12]}{\langle 14 \rangle \langle 32 \rangle \langle 34 \rangle^3} \end{aligned} \right)$$

* KLT relationship:

$$\begin{aligned} M_4(1234) &= \left(\frac{\langle 12 \rangle [12] \langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle} \right) \left(\frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 24 \rangle \langle 43 \rangle \langle 31 \rangle} \right) \\ &= \frac{\langle 12 \rangle^6 [12]}{\langle 23 \rangle \langle 34 \rangle \langle 41 \rangle \langle 24 \rangle \langle 43 \rangle \langle 31 \rangle} \\ &= \frac{-\langle 12 \rangle^6 [12]}{\langle 23 \rangle \langle 34 \rangle^2 \langle 14 \rangle \langle 13 \rangle \langle 24 \rangle} \end{aligned}$$

Last Attempt: $M_4(--++)$

$$\begin{aligned} M_4(--++) &= A_3[\hat{1}^- \hat{p}_I^+ 4^+] \left(\frac{1}{p_{14}^-} \right) A_3[-\hat{p}_I^- \hat{z}^- 3^+] \text{ (a)} \\ &+ A_3[\hat{1}^- \hat{p}_I^- 4^+] \frac{1}{p_{14}^+} A_3[-\hat{p}_I^+ \hat{z}^- 3^+] \text{ (b)} \end{aligned}$$

* Linearized gravity:

$$\begin{aligned} R_{\mu\nu}^L &= -\frac{1}{2} \square h_{\mu\nu} - \partial^\rho (\partial_\mu h_{\rho\nu} + \partial_\nu h_{\rho\mu}) + \partial_\mu \partial_\nu h \\ R^L &= -\frac{1}{2} \square h - 2 \partial_\rho \partial_\mu h^{\rho\mu} + \square h \end{aligned}$$

$\Rightarrow$ We have here to go to (at least) $\mathcal{O}(k^4)$

$$S = \int d^4x [k_0^2 h + k_1 h^2 \partial^2 h + k_2 h^3 \partial^2 h + k_3 \partial^2 h + \dots]$$

De Donder Gauge: $\partial^\mu h_{\mu\nu} = \frac{1}{2} \partial_\nu h$

$$h \partial^2 h \rightarrow \frac{1}{2} h_{\mu\nu} \square h^{\mu\nu} + \frac{1}{6} h_{\mu\nu}^2 \square h_{\mu\nu}$$

$$V_3(p_1 p_2 p_3) = p_1^{\mu_1} p_2^{\mu_2} \eta_{\mu_1 \nu_2} \eta_{\mu_2 \nu_1}$$

$$M_4(1^- 2^- 3^+) = \langle 12 \rangle^6 \quad A_3(1^- 2^- 3^+)^2$$

$$\begin{aligned} \mu_3 [1^+ 2^+ 3^-] &= \frac{[12]^6}{[23]^2 [31]^2} = A_3 [1^+ 2^+ 3^-]^2 \\ \mu_4 [1^- 2^- 3^+ 4^+] &= \frac{[12]^6 [12]}{[13][14][23][24][34]^2} = \frac{[12]^4 [34]^4}{864} \end{aligned}$$

(a) $\mu_3 [\hat{1} - \hat{p}^+ 4^+] \frac{1}{p_{14}^2} \mu_3 [-\hat{p}^- 2^- 3^+]$

$$= \frac{[1\hat{p}]^6}{[1\hat{p}]^2 [14]^2} \frac{1}{p_{14}^2} \left(\frac{[1\hat{p}]^6}{[23\hat{p}]^2 [3\hat{p}]^2} \right) = 0$$

$$\mu[4^- +] = 0$$

(b) Non-zero:
 $4 \leftrightarrow 3$ flip

$$\begin{aligned} &A[\hat{1} - \hat{p}^+ 3^+] \frac{1}{p_{13}^2} A_3 [-\hat{p}^- 2^- 4^+] \\ &+ A_3 [\hat{1} - \hat{p}^- 3^+] \frac{1}{p_{13}^2} A_3 [-\hat{p}^+ 2^- 4^+] \end{aligned}$$

Then

$$\mu_4 [- - + +] = \mu_3 [1 - \hat{p}^+ 4^+] \frac{1}{p_{14}^2} \mu_3 [-\hat{p}^- 2^- 3^+] + \mu_3 [1 - \hat{p}^+ 3^+] \frac{1}{p_{13}^2} \mu_3 [-\hat{p}^- 2^- 4^+] \quad (b)$$

(b) $\frac{[1\hat{p}]^6}{[1\hat{p}]^2 [13]^2} \frac{1}{p_{13}^2} \frac{[1\hat{p}]^6}{[24]^2 [3\hat{p}]^2}$

$$\begin{aligned} &= \frac{[3\hat{p}][\hat{p}|2]}{[34][4\hat{p}]} \quad \text{okay} \\ &= \frac{[34]^6 [24]^6}{[1\hat{p}]^2 [13]^2 [24]^2 [3\hat{p}]^2} \end{aligned}$$

$$\begin{aligned} &\mu_3 [1 - \hat{p}^- 3^+] \frac{1}{p_{13}^2} \mu_3 [-\hat{p}^+ 2^- 3^+] \\ &= \frac{[1\hat{p}]^6}{[1\hat{p}]^2 [13]^2} \frac{1}{p_{13}^2} \frac{[1\hat{p}]^6}{[24]^2 [3\hat{p}]^2} \end{aligned}$$

$$[\hat{1}\hat{2}][\hat{1}\hat{4}]^2 [\hat{1}\hat{3}] / [\hat{1}\hat{2}]$$

* Both $\mu_3[\hat{p}3]$ contribute are 0 zero
 $\mu_3[1-\hat{p}+4+] \frac{1}{p_{14}^2} \mu_3[-\hat{p}-2-3+]$

$$\mu_4 = \mu_3[1-\hat{p}_{14}-4+] \frac{1}{p_{14}^2} \mu_3[-\hat{p}_{14}^+ 2-3+] \textcircled{a}$$

$$\text{X} + \mu_3[1-\hat{p}_{14}^+ 4+] \frac{1}{p_{14}^2} \mu_3[-\hat{p}_{14}^- 2-3+] \textcircled{b}$$

$$+ \mu_3[1-\hat{p}_{13}^+ 3+] \frac{1}{p_{13}^2} \mu_3[-\hat{p}_{13}^+ 2-4+] \textcircled{c}$$

$$\text{X} + \mu_3[1-\hat{p}_{13}^* 3+] \frac{1}{p_{13}^2} \mu_3[-\hat{p}_{13}^- 2-4+] \textcircled{d}$$

$$\hat{p}_{14} : \hat{p}_{14}^2 = \langle \hat{1}4 \rangle [\hat{1}4] = 0 \quad \underline{\underline{\text{BRO}}}$$

$$\Rightarrow [\hat{1}4] = 0$$

$$\hat{p}_{13} : \hat{p}_{13}^2 = \langle \hat{1}3 \rangle [\hat{1}3] = 0 \Rightarrow [\hat{1}3] = 0$$

$$\textcircled{a} : \frac{\langle \hat{1}\hat{p}_{14} \rangle^6}{\langle 14 \rangle^2 \langle 1\hat{p} \rangle^2} \frac{1}{p_{14}^2} \frac{[\hat{p}3]^6}{[23]^2 [2\hat{p}]^2}$$

⑤ = Cannot have

$$\textcircled{c} \frac{\langle \hat{1}\hat{p} \rangle^6}{\langle 31 \rangle^2 \langle 3\hat{p} \rangle^2} \frac{1}{p_{13}^2} \frac{[\hat{p}4]^6}{[23]^2 [3\hat{p}]^2}$$

$$\langle \hat{1}\hat{p} \rangle [\hat{p}3] = \langle 14 \rangle [43]$$

$$\textcircled{a} \hookrightarrow \left(\frac{\langle 14 \rangle^6 [43]^6}{\langle 14 \rangle^2 \langle 1\hat{p} \rangle^2} \right) \left(\frac{1}{p_{14}^2} \right) \frac{1}{[23]^2 [2\hat{p}]^2}$$

$$\langle \hat{1}\hat{p} \rangle [\hat{p}2] = \langle 14 \rangle [24]$$

$$\frac{\langle 14 \rangle^6 [34]^6}{[24]^2 [23]^2 [14]}$$

$$= \frac{\langle 14 \rangle^4 [24]^2 [23]^2 \langle 14 \rangle [14]}{[24]^2 [23]^2 [14]}$$

$$= \frac{\langle 14 \rangle [34]^6}{[24]^2 [23]^2 [14]}$$

$$\langle 14 \rangle [43] = -\langle 12 \rangle [23]$$

$$= \langle 12 \rangle [34]^5$$

$$[24]^2 [23]^2 [14]$$

$$\frac{[34]}{[24]} = \frac{[43]}{[42]} = \frac{\langle 21 \rangle}{\langle 31 \rangle}$$

$$\rightarrow \frac{\langle 12 \rangle}{\langle 13 \rangle^2} \left(\frac{\langle 21 \rangle^2}{\langle 13 \rangle^2} \right) \frac{[34]^3}{[23][14]}$$

$$= \frac{\langle 12 \rangle^3}{\langle 13 \rangle^2} \left(\frac{[34]}{[23]} \right) \left(\frac{[34]}{[14]} \right) \left(\frac{[34]}{[14]} \right)$$

$$= \frac{[34]}{[32]} = \frac{\langle 21 \rangle}{\langle 41 \rangle} \quad \bigg/ \quad \frac{[43]}{[41]} = \frac{\langle 12 \rangle}{\langle 32 \rangle}$$

$$\rightarrow \frac{\langle 12 \rangle^5 [34]}{\langle 14 \rangle \langle 23 \rangle \langle 13 \rangle^2}$$

$$\circ [34] \langle 41 \rangle + [32] \langle 21 \rangle + [31] \langle 11 \rangle = 0$$

$$[34] \langle 42 \rangle = -[31] \langle 12 \rangle$$

$$\frac{\langle 12 \rangle^6 [13]}{\langle 42 \rangle \langle 14 \rangle \langle 23 \rangle \langle 13 \rangle^2} = \underline{P_{14}}$$

② $\frac{\langle 1\hat{p} \rangle^6}{\langle 31 \rangle^2 \langle 1\hat{p} \rangle^2} \quad \frac{1}{P_{13}^2} \frac{[\hat{p}4]^6}{[\hat{2}\hat{p}]^2 [\hat{2}4]^2} \quad P_{13}$

$$\left(\begin{array}{l} \langle 1\hat{p} \rangle [\hat{p}4] = \langle 13 \rangle [34] \\ \langle 3\hat{p} \rangle [\hat{p}2] = \langle 31 \rangle [12] \end{array} \right)$$

$$\rightarrow \frac{\langle 13 \rangle^6 [34]^6}{\langle 13 \rangle^4 [12]^2 \langle 13 \rangle [13] [24]^2}$$

$$= \frac{\langle 13 \rangle [34]^6}{[12]^2 [13] [24]^2} = \frac{\langle 12 \rangle^2 [34]^4}{\langle 13 \rangle [12]^2 [13]}$$

$$\circ [34] = -\langle 12 \rangle$$

$$\begin{aligned}
 & \left(\begin{array}{l} [31] \\ \langle 42 \rangle \end{array} \right) = \frac{\langle 42 \rangle^3 [34]^3}{\langle 24 \rangle [12]^2} = \frac{\langle 12 \rangle^3 [34]^3}{[13] \langle 23 \rangle [12]} \\
 & = \frac{\langle 12 \rangle^4 [34]^2}{\langle 24 \rangle \langle 23 \rangle [12]}
 \end{aligned}$$

$$\begin{aligned}
 & \left(\begin{array}{l} [34] \langle 42 \rangle + [31] \langle 12 \rangle = 0 \\ [34] = - \frac{[13] \langle 12 \rangle}{\langle 24 \rangle} \end{array} \right) \\
 & \Rightarrow \frac{\langle 12 \rangle^6 [13]^2}{\langle 24 \rangle^3 \langle 23 \rangle [12]}
 \end{aligned}$$

$$\begin{aligned}
 & \left(\begin{array}{l} [13] \\ [12] \end{array} \right) = - \frac{\langle 24 \rangle}{\langle 34 \rangle} \\
 & \Rightarrow \frac{\langle 12 \rangle^6 [13]}{\langle 24 \rangle^2 \langle 34 \rangle \langle 23 \rangle}
 \end{aligned}$$

$$= \frac{\langle 12 \rangle^6 [13]}{\langle 24 \rangle^2 \langle 34 \rangle} \left(\frac{1}{\langle 24 \rangle^2 \langle 34 \rangle} + \frac{1}{\langle 24 \rangle \langle 14 \rangle \langle 23 \rangle \langle 13 \rangle^2} \right)$$

$$= \frac{\langle 12 \rangle^6 [13]}{\langle 24 \rangle \langle 14 \rangle \langle 23 \rangle \langle 13 \rangle^2}$$

$$[13] \langle 31 \rangle + [12] \langle 21 \rangle + [14] \langle 41 \rangle + \dots = 0$$

$$[13] = - \frac{[12] \langle 24 \rangle}{\langle 34 \rangle}$$

$$= \frac{\langle 12 \rangle^6 [12]}{\langle 34 \rangle \langle 14 \rangle \langle 23 \rangle \langle 13 \rangle^2} : P_{14} = \text{u-channel}$$

$$\text{t-channel} : \hat{P}_{13}$$

$$\frac{\langle \hat{1} \hat{p} \rangle^6}{\langle 13 \rangle \langle 3 \hat{p} \rangle^2} \left(\frac{1}{\hat{P}_{13}^2} \right) \left(\frac{[\hat{P}_{14}]^6}{[24]^2 [2 \hat{p}]^2} \right)$$

$$\langle 1 \hat{4} \rangle [\hat{4} 1] = \langle 12 \rangle [2 4]$$

$$\begin{aligned}
 & \langle 13 \rangle^6 [34]^6 \\
 & \frac{\langle 13 \rangle^2 [24]^2 \langle 13 \rangle [13] \langle 31 \rangle^2 [21]^2}{\langle 13 \rangle [34]^6} \\
 & = \frac{\langle 21 \rangle^2}{\langle 31 \rangle^2} \left(\frac{\langle 13 \rangle [34]^4}{[13] [12]^2} \right) \\
 & = \frac{\langle 12 \rangle^2 [34]^3}{\langle 13 \rangle [12]^2} \left(\frac{[34]}{[13]} \right)
 \end{aligned}$$

$$\left(\frac{[34] \langle 12 \rangle}{[31]} = \langle 42 \rangle \right)$$

$$\frac{\langle 12 \rangle^3 [34]^3}{\langle 13 \rangle \langle 42 \rangle [12]^2} = \frac{\langle 12 \rangle^3 \langle 12 \rangle^3 [13]^3}{\langle 24 \rangle^4 \langle 13 \rangle [12]^2}$$

$$\begin{aligned}
 \frac{[13]}{[12]} &= -\frac{\langle 24 \rangle}{\langle 34 \rangle} \\
 & \Rightarrow \langle 12 \rangle^6 [13] \\
 & \frac{\langle 34 \rangle^2 \langle 24 \rangle^2 \langle 13 \rangle}{\langle 13 \rangle \langle 34 \rangle^3 \langle 24 \rangle^2} \\
 & = -\langle 12 \rangle^6 [12]
 \end{aligned}$$

$$\begin{aligned}
 P_{14} &= A_3 [1^- p_{14}^+ u^+] \frac{1}{p_{14}^2} A_3 [-\hat{p}^+ 2^- 3^+] \\
 &= \frac{\langle 1\hat{p} \rangle^6}{\langle 14 \rangle^2 \langle 1\hat{p} \rangle^2} \frac{1}{p_{14}^2} \frac{[\hat{p} 2]^2}{[3\hat{p}]^2 [32]^2}
 \end{aligned}$$

$$\begin{aligned}
 \underline{M_{u \text{ old}}} &: M_4 [1^- 2^- 3^+ u^+] \quad \hat{p}_T = \hat{z} + 3 \\
 M_3 [1^- p^+ u^+] &= \frac{[\hat{p} 4]^6}{[\hat{1}\hat{p}]^2 [41]^2}
 \end{aligned}$$

$$\hat{p}_{23}^2 = 0 \Rightarrow \langle \hat{z} 3 \rangle = 0$$

$$M_3 (-\hat{p}^- 2^- 3^+) = 0$$

$$M_4 (- - + +) = M_3 (\hat{z}^- \hat{p}^- u^+) \frac{1}{\hat{p}_{23}^2} M_3 (-\hat{p}^+ 2^- 3^+)$$

$$= \frac{\langle \hat{1}\hat{2} \rangle^6}{\langle 14 \rangle^2 \langle \hat{1}\hat{4} \rangle^2} \left(\frac{1}{\langle 23 \rangle \langle 2\bar{3} \rangle} \right) \frac{[\hat{3}\hat{1}]^2}{[\hat{2}\hat{1}]^2 [\hat{2}\bar{3}]^2}$$

$$= \frac{\langle 12 \rangle^6 [\hat{2}\bar{3}]^6}{\langle 14 \rangle^2 \langle \hat{1}\hat{4} \rangle^2 [\hat{2}\hat{1}]^2 [\hat{2}\bar{3}]^2 \langle 23 \rangle [\hat{2}\bar{3}]}$$

$$\left(\begin{aligned} & [\hat{2}\hat{1}] \langle \hat{1}\hat{4} \rangle = [\hat{2}\bar{3}] \langle \bar{1}\bar{4} \rangle \\ & \frac{\langle 12 \rangle^6 [\hat{2}\bar{3}]^6}{\langle 14 \rangle^2 [\hat{2}\bar{3}]^2 \langle \bar{3}\bar{4} \rangle^2 [\hat{2}\bar{3}]^3 \langle 23 \rangle} \end{aligned} \right)$$

$$= \frac{\langle 12 \rangle^6 [\hat{2}\bar{3}]}{\langle 14 \rangle^2 \langle \bar{3}\bar{4} \rangle^2 \langle 23 \rangle}$$

$$0 = \frac{[\hat{2}\bar{3}] \langle \bar{3}\bar{1} \rangle}{\langle 14 \rangle} + \frac{[\hat{2}\bar{1}] \langle \bar{1}\bar{1} \rangle}{\langle 23 \rangle} + \frac{[\hat{2}\bar{4}] \langle \bar{4}\bar{1} \rangle}{\langle 23 \rangle \langle \bar{3}\bar{4} \rangle} + \frac{[\hat{2}\bar{1}] \langle \bar{1}\bar{4} \rangle}{\langle 23 \rangle}$$

$$\frac{[\hat{2}\bar{3}]}{\langle 14 \rangle} = \frac{[\hat{1}\bar{2}]}{\langle \bar{3}\bar{4} \rangle}$$

$$\left(\frac{\langle 12 \rangle^7 [\hat{1}\bar{2}]}{\langle 14 \rangle \langle \bar{3}\bar{4} \rangle^3 \langle 23 \rangle \langle \bar{1}\bar{2} \rangle} \right)$$

$$\langle \bar{4}\bar{1}\bar{2} \rangle \langle 23 \rangle + \langle \bar{4}\bar{1}\bar{2} \rangle \langle \bar{3}\bar{1} \rangle + \langle \bar{4}\bar{1}\bar{3} \rangle \langle \bar{1}\bar{2} \rangle$$

$$\langle \bar{4}\bar{2} \rangle \langle \bar{3}\bar{1} \rangle + \langle \bar{4}\bar{3} \rangle \langle \bar{1}\bar{2} \rangle = 0$$

$$\langle \bar{4}\bar{2} \rangle \langle \bar{3}\bar{1} \rangle = \langle \bar{3}\bar{4} \rangle \langle \bar{1}\bar{2} \rangle$$

$$\frac{\langle 12 \rangle^2 [\hat{1}\bar{2}]}{\langle 14 \rangle \langle \bar{3}\bar{4} \rangle^2 \langle 23 \rangle \langle \bar{1}\bar{3} \rangle \langle \hat{2}\bar{4} \rangle}$$

$$\frac{\langle 12 \rangle^6 [\hat{1}\bar{2}]}{\langle \bar{3}\bar{4} \rangle^3} \left(\frac{1}{\langle 14 \rangle \langle 23 \rangle} - \frac{1}{\langle \bar{1}\bar{3} \rangle \langle \bar{2}\bar{4} \rangle} \right)$$

$$\frac{\langle 12 \rangle^6 [\hat{1}\bar{2}]}{\langle \bar{3}\bar{4} \rangle^3} \left(\frac{\langle \bar{1}\bar{3} \rangle \langle \bar{2}\bar{4} \rangle - \langle 14 \rangle \langle 23 \rangle}{\langle 14 \rangle \langle 23 \rangle \langle \bar{1}\bar{3} \rangle \langle \bar{2}\bar{4} \rangle} \right)$$

$$= \frac{\langle 12 \rangle^6 [\hat{1}\bar{2}]}{\langle \bar{3}\bar{4} \rangle^3 \langle \bar{1}\bar{3} \rangle \langle \bar{2}\bar{4} \rangle}$$

$$= \frac{\langle 34 \rangle^2 \left(\langle 14 \rangle \langle 23 \rangle \langle 13 \rangle \langle 24 \rangle \right)}{\langle 12 \rangle^7 [12]} \\ \langle 34 \rangle^2 \langle 14 \rangle \langle 13 \rangle \langle 23 \rangle \langle 24 \rangle$$

$$* \mu_3(\hat{I} - \hat{P} - 3+) \frac{1}{\hat{P}_n^2} \mu_3(-\hat{P} + 2 - 4^+)$$